

Hessian Geometry, Local Rates, and Acceleration of Sinkhorn

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Abstract

This note isolates one conditional-expectation operator that controls three aspects of entropic optimal transport: dual curvature, the local convergence of Sinkhorn's algorithm, and semi-dual conditioning. At the optimal coupling π_ϵ^* , write $T_\epsilon := T_{\pi_\epsilon^*}$. The positive full-dual curvature operator is $\epsilon^{-1} \begin{pmatrix} I & T_\epsilon \\ T_\epsilon^* & I \end{pmatrix}$, the Jacobian of one complete Sinkhorn cycle is $T_\epsilon^* T_\epsilon$, and exact elimination of one potential gives the semi-dual Hessian $\epsilon^{-1}(I - T_\epsilon^* T_\epsilon)$. This common structure yields the local rate, strict-gap criteria, small- ϵ asymptotics, and explicit parameters for extrapolation and momentum. It also explains why variable projection and quasi-Newton methods are natural alternatives to long scaling iterations.

1 Dual Hessian and conditional expectation

The correct geometry for the dual potentials is not an unweighted Euclidean space: it is the product of the two marginal L^2 spaces. In this geometry, the Hessian has an especially simple block form.

Entropic primal and dual. Let $\mathcal{X}$ and $\mathcal{Y}$ be standard Borel spaces, let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and let $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{+\infty\}$ be measurable. The KL-regularized transport value is

$$\mathcal{L}_c^\epsilon(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \left\{ \int c \, d\pi + \epsilon \, \text{KL}(\pi \parallel \alpha \otimes \beta) \right\}. \quad (1)$$

This is the entropy-regularized transport problem introduced computationally by Cuturi [3]. Its concave dual objective is

$$\mathcal{D}_\epsilon(f, g) := \int_{\mathcal{X}} f \, d\alpha + \int_{\mathcal{Y}} g \, d\beta - \epsilon \int_{\mathcal{X} \times \mathcal{Y}} \left[\exp \left(\frac{f(x) + g(y) - c(x, y)}{\epsilon} \right) - 1 \right] d\alpha(x) d\beta(y) \quad (2)$$

and $\mathcal{L}_c^\epsilon(\alpha, \beta) = \sup_{f, g} \mathcal{D}_\epsilon(f, g)$ under the standard duality assumptions. For any potentials for which the integral is finite, denote their Gibbs measure by

$$d\pi_{f, g}(x, y) = \exp \left(\frac{f(x) + g(y) - c(x, y)}{\epsilon} \right) d\alpha(x) d\beta(y) \quad (3)$$

and write $(\pi_{f, g})_1, (\pi_{f, g})_2$ for its marginals. The first variation is

$$D\mathcal{D}_\epsilon(f, g)[h, k] = \int_{\mathcal{X}} h \, d(\alpha - (\pi_{f, g})_1) + \int_{\mathcal{Y}} k \, d(\beta - (\pi_{f, g})_2). \quad (4)$$

Consequently, $\pi_{f, g} \in \mathcal{U}(\alpha, \beta)$ exactly when (f, g) is stationary. Since $\mathcal{D}_\epsilon$ is concave, every such pair is a dual maximizer. Assume henceforth that a maximizer (f^*, g^*) exists, and set

$$\pi_\epsilon^* := \pi_{f^*, g^*} \in \mathcal{U}(\alpha, \beta). \quad (5)$$

The gauge transformation $(f, g) \mapsto (f + s, g - s)$ leaves (2) unchanged. We isolate the nonconstant correlation modes through the centered spaces

$$L_0^2(\alpha) := \left\{ h : \int h \, d\alpha = 0 \right\}, \quad L_0^2(\beta) := \left\{ k : \int k \, d\beta = 0 \right\},$$

endowed with their usual weighted norms. We write

$$\text{cent}_\alpha h := h - \int h \, d\alpha, \quad \text{cent}_\beta k := k - \int k \, d\beta \quad (6)$$

for the orthogonal centering projectors. The full quotient by the gauge also retains the common-constant direction $(\mathbf{1}, \mathbf{1})$; its curvature will be recorded separately.

Coupling operator. The operator governing the local theory is meaningful for every coupling, independently of entropic optimality.

Definition 1.1 (Conditional operator and maximal correlation). For $\pi \in \mathcal{U}(\alpha, \beta)$, disintegrate π with respect to each marginal and define

$$T_\pi : L^2(\beta) \longrightarrow L^2(\alpha), \quad T_\pi k(x) := \mathbb{E}_\pi[k(Y) \mid X = x], \quad (7)$$

$$T_\pi^* : L^2(\alpha) \longrightarrow L^2(\beta), \quad T_\pi^* h(y) := \mathbb{E}_\pi[h(X) \mid Y = y]. \quad (8)$$

Its centered maximal-correlation coefficient is

$$\begin{aligned} \sigma(\pi) &:= \left\| T_\pi : L_0^2(\beta) \rightarrow L_0^2(\alpha) \right\|_{\text{op}} \\ &= \sup_{\substack{h \in L_0^2(\alpha), \, k \in L_0^2(\beta) \\ \|h\|_\alpha = \|k\|_\beta = 1}} \int_{\mathcal{X} \times \mathcal{Y}} h(x)k(y) \, d\pi(x, y). \end{aligned} \quad (9)$$

For the entropic optimizer (5), set

$$T_\epsilon := T_{\pi_\epsilon^*}, \quad \sigma_\epsilon := \sigma(\pi_\epsilon^*), \quad \gamma_\epsilon := 1 - \sigma_\epsilon^2. \quad (10)$$

The adjoint notation is literal: $\langle h, T_\pi k \rangle_\alpha = \langle T_\pi^* h, k \rangle_\beta$. Conditional Jensen gives $\|T_\pi\|_{\text{op}} \leq 1$, while $T_\pi \mathbf{1} = \mathbf{1}$ and $T_\pi^* \mathbf{1} = \mathbf{1}$. Hence T_π and T_π^* preserve centered functions, $\sigma(\pi) \in [0, 1]$, and γ_ϵ is the spectral gap of $T_\epsilon^* T_\epsilon$ on $L_0^2(\beta)$.

Proposition 1.2 (Dual Hessian at arbitrary and optimal potentials). *For arbitrary potentials (f, g) and directions $(h, k), (h', k')$ for which the expressions are finite,*

$$-\epsilon D^2 \mathcal{D}_\epsilon(f, g)[(h, k), (h', k')] = \int_{\mathcal{X} \times \mathcal{Y}} (h(x) + k(y))(h'(x) + k'(y)) \, d\pi_{f, g}(x, y). \quad (11)$$

At an optimal pair (f^, g^*) , the positive curvature operator on $L^2(\alpha) \oplus L^2(\beta)$ is*

$$H_{\text{dual}} = -\nabla^2 \mathcal{D}_\epsilon(f^*, g^*) = \frac{1}{\epsilon} \begin{pmatrix} I & T_{\pi_\epsilon^*} \\ T_{\pi_\epsilon^*}^* & I \end{pmatrix}. \quad (12)$$

On $L_0^2(\alpha) \oplus L_0^2(\beta)$,

$$\frac{1 - \sigma_\epsilon}{\epsilon} I \preceq H_{\text{dual}} \preceq \frac{1 + \sigma_\epsilon}{\epsilon} I, \quad \kappa_{\text{dual}, 0} = \frac{1 + \sigma_\epsilon}{1 - \sigma_\epsilon}. \quad (13)$$

On the full space modulo the gauge direction $(\mathbf{1}, -\mathbf{1})$, the additional eigenvalue is $2/\epsilon$, and hence

$$\kappa_{\text{dual}} = \frac{2}{1 - \sigma_\epsilon}. \quad (14)$$

Proof. Twice differentiating the exponential term in $-\mathcal{D}_\epsilon$ gives (11). At the optimum, $\pi_{f^*, g^*} = \pi_\epsilon^*$ has marginals (α, β) . The two diagonal terms in (11) are therefore the $L^2(\alpha)$ and $L^2(\beta)$ inner products, while the cross term is $\langle h, T_{\pi_\epsilon^*} k \rangle_\alpha$. This proves (12). On a centered singular pair $T_\epsilon v = su$, $T_\epsilon^* u = sv$, the two vectors $(u, \pm v)$ have eigenvalues $(1 \pm s)/\epsilon$. Taking $s = \sigma_\epsilon$ proves (13). The gauge vector $(\mathbf{1}, -\mathbf{1})$ has eigenvalue zero, whereas $(\mathbf{1}, \mathbf{1})$ has eigenvalue $2/\epsilon$, proving (14). $\square$

Thus optimality is not needed for the second-variation identity (11). It is needed for the clean conditional-expectation form (12): away from stationarity, $\pi_{f, g}$ is generally not a coupling, the diagonal blocks are multiplication by its marginal densities relative to α and β , and the off-diagonal block is an unnormalized integral operator rather than T_π from Definition 1.1.

For discrete marginals $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, with optimal coupling matrix P_ϵ^* , the same operators read

$$T_{\pi_\epsilon^*} k = D_a^{-1} P_\epsilon^* k, \quad T_{\pi_\epsilon^*}^* h = D_b^{-1} (P_\epsilon^*)^\top h,$$

where $D_a = \text{diag}(a)$ and $D_b = \text{diag}(b)$. These formulas are adjoint in the a - and b -weighted inner products, not in the unweighted Euclidean ones.

2 Sinkhorn Jacobian and local rate

A full Sinkhorn cycle is a nonlinear map on one potential. Its linearization at the fixed point is exactly the two-step conditional expectation $T_\epsilon^* T_\epsilon$, so the local convergence question reduces to a Markov spectral gap.

Soft transforms and the cycle map. Exact block maximization of $\mathcal{D}_\epsilon$ is written with the continuous soft c -transforms

$$g^{\bar{c}, \epsilon}(x) := -\epsilon \log \int_Y \exp \left(\frac{g(y) - c(x, y)}{\epsilon} \right) d\beta(y), \quad (15)$$

$$f^{c, \epsilon}(y) := -\epsilon \log \int_X \exp \left(\frac{f(x) - c(x, y)}{\epsilon} \right) d\alpha(x). \quad (16)$$

One complete cycle on the target potential is therefore

$$\mathcal{S}_\epsilon(g) := (g^{\bar{c}, \epsilon})^{c, \epsilon}, \quad g^{(\ell+1)} = \mathcal{S}_\epsilon(g^{(\ell)}). \quad (17)$$

The opposite ordering acts on $L^2(\alpha)$ and has the same nonzero spectrum.

Proposition 2.1 (Jacobian and exact local Sinkhorn rate). *Assume that the two soft transforms are twice differentiable near the optimum. Then*

$$D[g \mapsto g^{\bar{c}, \epsilon}](g^*) = -T_\epsilon, \quad D[f \mapsto f^{c, \epsilon}](f^*) = -T_\epsilon^*, \quad D\mathcal{S}_\epsilon(g^*) = T_\epsilon^* T_\epsilon. \quad (18)$$

Consequently, for the centered error $e^{(\ell)} = \text{cent}_\beta(g^{(\ell)} - g^*)$ of any Sinkhorn sequence converging to g^* modulo constants,

$$e^{(\ell+1)} = T_\epsilon^* T_\epsilon e^{(\ell)} + o(\|e^{(\ell)}\|_\beta), \quad \limsup_{\ell \rightarrow \infty} \frac{\|e^{(\ell+1)}\|_\beta}{\|e^{(\ell)}\|_\beta} \leq \sigma_\epsilon^2. \quad (19)$$

If σ_ϵ^2 is an isolated eigenvalue and the asymptotic error has a nonzero projection onto its eigenspace, the ratio converges to σ_ϵ^2 . Moreover,

$$\gamma_\epsilon = \inf_{\substack{k \in L_0^2(\beta) \\ \|k\|_\beta = 1}} \mathbb{E}_{\pi_\epsilon^*} [\text{Var}(k(Y) \mid X)]. \quad (20)$$

Proof. Differentiating the log-partition maps (15)–(16) at the optimal Gibbs coupling gives the first two identities in (18); the chain rule gives the third. Taylor expansion proves (19). Finally, for centered k ,

$$\langle k, (I - T_\epsilon^* T_\epsilon) k \rangle_\beta = \|k\|_\beta^2 - \|T_\epsilon k\|_\alpha^2 = \mathbb{E}_{\pi_\epsilon^*}[\text{Var}(k(Y) \mid X)].$$

Taking the infimum over the unit sphere gives (20). $\square$

The reverse cycle $f \mapsto (f^{c, \epsilon})^{\bar{c}, \epsilon}$ has Jacobian $T_\epsilon T_\epsilon^*$ on $L_0^2(\alpha)$. Since $T_\epsilon^* T_\epsilon$ and $T_\epsilon T_\epsilon^*$ have the same nonzero spectrum, either orientation has the same local rate. In finite spaces, this recovers the classical second-singular-value formula for matrix scaling [12, 13].

When is the rate strictly smaller than one? Positivity of the Gibbs kernel is enough in finite dimension but not on an arbitrary infinite space: centered approximate modes may retain correlation arbitrarily close to one. The following input-level bound is simple and explicit.

Define the cost oscillation modulo separable terms by

$$\Omega_c := \inf_{\xi, \zeta} \text{ess osc}_{\alpha \otimes \beta} (c(x, y) - \xi(x) - \zeta(y)). \quad (21)$$

Here the infimum is over measurable real-valued functions ξ on $\mathcal{X}$ and ζ on $\mathcal{Y}$. For compact supports and continuous c , also define the invariant four-point oscillation

$$\Delta_c := \sup_{x, x', y, y'} [c(x, y') + c(x', y) - c(x, y) - c(x', y')], \quad (22)$$

where the supremum is over the product of the two supports.

Proposition 2.2 (Explicit strict-gap criteria). *If $\Omega_c < \infty$, then*

$$\sigma_\epsilon \leq 1 - e^{-2\Omega_c/\epsilon} < 1, \quad \sigma_\epsilon^2 \leq (1 - e^{-2\Omega_c/\epsilon})^2. \quad (23)$$

If the supports are compact and c is continuous, then the projective bound also holds:

$$\sigma_\epsilon \leq \Lambda_\epsilon := \tanh\left(\frac{\Delta_c}{4\epsilon}\right) < 1. \quad (24)$$

Proof. Adding $\xi(x) + \zeta(y)$ to c only changes the dual potentials. For any $\eta > 0$, choose such terms so that the resulting cost has essential oscillation at most $\Omega_c + \eta$. Write $k_\epsilon = e^{-c/\epsilon}$, $m = \text{ess inf } k_\epsilon$, $M = \text{ess sup } k_\epsilon$, and $d\pi_\epsilon^* = u(x)k_\epsilon(x, y)v(y) d\alpha d\beta$. The two marginal equations give

$$\frac{1}{MV} \leq u \leq \frac{1}{mV}, \quad \frac{1}{MU} \leq v \leq \frac{1}{mU}, \quad \frac{1}{M} \leq UV \leq \frac{1}{m},$$

where $U = \int u d\alpha$ and $V = \int v d\beta$. Hence the density $r = d\pi_\epsilon^*/d(\alpha \otimes \beta)$ satisfies $r \geq (m/M)^2 \geq e^{-2(\Omega_c + \eta)/\epsilon} =: \delta_\eta$. Decomposing $r = \delta_\eta + (1 - \delta_\eta)\tilde{r}$ shows that, on centered functions, $T_\epsilon = (1 - \delta_\eta)\tilde{T}_\epsilon$ with $\|\tilde{T}_\epsilon\|_{\text{op}} \leq 1$. Letting $\eta \downarrow 0$ proves (23).

Under compactness, the conditional kernel is continuous and T_ϵ is compact. Birkhoff's theorem bounds both half-cycles in Hilbert's projective metric by Λ_ϵ [7]. If $\sigma_\epsilon > 0$, compactness provides a continuous centered singular pair $T_\epsilon v = \sigma_\epsilon u$, $T_\epsilon^* u = \sigma_\epsilon v$. Differentiating the two projective bounds gives $\sigma_\epsilon \text{osc}(u) \leq \Lambda_\epsilon \text{osc}(v)$ and the reverse inequality with u, v exchanged. Multiplication yields $\sigma_\epsilon^2 \leq \Lambda_\epsilon^2$; the case $\sigma_\epsilon = 0$ is immediate. $\square$

On unbounded spaces, cost growth must be balanced against marginal tails. Conforti, Durmus, and Greco prove quantitative contraction for quadratic costs under asymptotically positive log-concavity profiles [2]; Eckstein obtains weighted Hilbert contraction under compatible cost-growth and tail assumptions [6]. These are useful input-level criteria, but their norms should not be identified automatically with the exact L^2 gap (20).

Why a linear small- ϵ gap is natural. In a smooth connected Monge regime, the conditional law of Y given $X = x$ has width $O(\sqrt{\epsilon})$ around the hard transport map. The conditional variance in (20) is therefore typically of order ϵ , leading to $\sigma_\epsilon^2 = 1 - C\epsilon + o(\epsilon)$.

A precise conditional statement is as follows. Assume that $\mathcal{X}$ and $\mathcal{Y}$ are smooth connected d -dimensional manifolds, that the marginals have positive smooth densities, and that hard transport is induced by a smooth diffeomorphism $\mathbf{M} : \mathcal{X} \rightarrow \mathcal{Y}$. Let g_0^* be the hard target potential and set

$$H(x) := D_{yy}^2(c(x, \cdot) - g_0^*)(\mathbf{M}(x)), \quad A(y) := H(\mathbf{M}^{-1}(y))^{-1}. \quad (25)$$

Suppose that H is uniformly positive and that the uniform Laplace expansion of the entropic conditionals permits passage to the first nonzero Rayleigh quotient, for instance through Mosco convergence of the rescaled forms and compact convergence of their resolvents. Define

$$\lambda_1(A, \beta) := \inf_{\substack{k \in \mathcal{H} \\ \int k^2 d\beta = 1}} \int_{\mathcal{Y}} \langle A(y) \nabla k(y), \nabla k(y) \rangle d\beta(y). \quad (26)$$

Uniform ellipticity and a Poincaré inequality for β ensure that $\lambda_1(A, \beta) > 0$.

Proposition 2.3 (Smooth small-regularization law). *Under the preceding assumptions,*

$$\gamma_\epsilon = \epsilon \lambda_1(A, \beta) + o(\epsilon), \quad \sigma_\epsilon^2 = 1 - \epsilon \lambda_1(A, \beta) + o(\epsilon), \quad \sigma_\epsilon = 1 - \frac{\epsilon}{2} \lambda_1(A, \beta) + o(\epsilon). \quad (27)$$

Proof. For the optimal coupling π_ϵ^* , the uniform Laplace expansion of $Y \mid X = x$ gives $\text{Cov}(Y \mid X = x) = \epsilon H(x)^{-1} + o(\epsilon)$. Hence, for smooth k ,

$$\frac{1}{\epsilon} \mathbb{E}_{\pi_\epsilon^*}[\text{Var}(k(Y) \mid X)] \longrightarrow \int_{\mathcal{X}} \langle H(x)^{-1} \nabla k(\mathbf{M}(x)), \nabla k(\mathbf{M}(x)) \rangle d\alpha(x).$$

Since $\mathbf{M}_\# \alpha = \beta$, the right-hand side is the form in (26). Spectral convergence permits passage to the first Rayleigh quotient in (20), giving the first expansion. The other two follow from $\gamma_\epsilon = 1 - \sigma_\epsilon^2$. $\square$

Example 2.4 (Exact Gaussian benchmark). For $\alpha = \beta = \mathcal{N}(0, 1)$ and $c(x, y) = \frac{1}{2}(x - y)^2$, the optimal coupling is Gaussian with correlation

$$r_\epsilon = \frac{\sqrt{\epsilon^2 + 4} - \epsilon}{2}. \quad (28)$$

Hermite polynomials diagonalize T_ϵ , so $\sigma_\epsilon = r_\epsilon$. Thus $\sigma_\epsilon = 1 - \epsilon/2 + O(\epsilon^2)$ and $\sigma_\epsilon^2 = 1 - \epsilon + O(\epsilon^2)$. This is Proposition 2.3 with $A = 1$ and $\lambda_1(A, \beta) = 1$.

Local L^2 versus global Hilbert contraction. For $\Delta_c > 0$,

$$1 - \Lambda_\epsilon^2 \sim 4e^{-\Delta_c/(2\epsilon)} \quad (\epsilon \downarrow 0). \quad (29)$$

This is exponentially more pessimistic than the smooth local law $\gamma_\epsilon \sim \epsilon \lambda_1(A, \beta)$. There is no contradiction. Hilbert's projective metric gives a global guarantee over the full positive cone and uses no smoothness of the iterates. The L^2 result is local to the optimal coupling and exploits the regular diffusive structure of the dominant modes. Chizat, Delalande, and Vaškevičius similarly obtain polynomial rather than universal exponential dependences under continuous regularity and log-concavity assumptions [1].

3 Extrapolation and momentum

Once the local spectrum is reduced to $\text{spec}(T_\epsilon^* T_\epsilon) \subset [0, \sigma_\epsilon^2]$, standard polynomial acceleration applies. Scalar extrapolation and momentum are two instances of the same stationary two-step iteration.

Two-parameter acceleration. For $\omega > 0$ and $\vartheta \geq 0$, consider

$$g^{(\ell+1)} = g^{(\ell)} + \omega(\mathcal{S}_\epsilon(g^{(\ell)}) - g^{(\ell)}) + \vartheta(g^{(\ell)} - g^{(\ell-1)}). \quad (30)$$

The choice $\vartheta = 0$ gives scalar extrapolation, while $(\omega, \vartheta) = (1, 0)$ is ordinary Sinkhorn. Locally, (30) is the heavy-ball iteration applied to $I - T_\epsilon^* T_\epsilon$ [16]. On a mode with $T_\epsilon^* T_\epsilon k = \lambda k$, write $a := 1 - \lambda \in [\gamma_\epsilon, 1]$; the scalar error satisfies

$$z_{\ell+1} = (1 - \omega a + \vartheta)z_\ell - \vartheta z_{\ell-1}. \quad (31)$$

Proposition 3.1 (Optimal extrapolation and momentum parameters). *Assume $\gamma_\epsilon > 0$.*

1. *Among iterations (30) with $\vartheta = 0$, the minimax choice and its worst local factor are*

$$\omega_{\text{ext}} = \frac{2}{1 + \gamma_\epsilon}, \quad r_{\text{ext}} = \frac{1 - \gamma_\epsilon}{1 + \gamma_\epsilon} = \frac{\sigma_\epsilon^2}{2 - \sigma_\epsilon^2}. \quad (32)$$

2. *Among stationary two-step iterations (30) with $\omega > 0$ and $\vartheta \geq 0$, the minimax parameters are*

$$\omega_{\text{mom}} = \frac{4}{(1 + \sqrt{\gamma_\epsilon})^2}, \quad \vartheta_{\text{mom}} = \left(\frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}} \right)^2. \quad (33)$$

Their worst local asymptotic factor is

$$r_{\text{mom}} = \frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}}. \quad (34)$$

Thus extrapolation doubles the small gap to first order, whereas momentum changes its scale from γ_ϵ to $\sqrt{\gamma_\epsilon}$.

Proof. If $\vartheta = 0$, the mode factor is $1 - \omega a$. Balancing its values at $a = \gamma_\epsilon$ and $a = 1$ gives (32).

For the two-step problem, the case $\gamma_\epsilon = 1$ is immediate. Assume $0 < \gamma_\epsilon < 1$ and suppose that both roots of $z^2 - (1 - \omega a + \vartheta)z + \vartheta$ have modulus at most $r < 1$ for every $a \in [\gamma_\epsilon, 1]$. Applying the elementary quadratic unit-disk criterion after rescaling the roots by r gives

$$0 \leq \vartheta \leq r^2, \quad -r - \frac{\vartheta}{r} \leq 1 - \omega a + \vartheta \leq r + \frac{\vartheta}{r}.$$

Applying the upper inequality at $a = \gamma_\epsilon$ and the lower one at $a = 1$, then using $\vartheta/r \leq r$, yields the necessary condition

$$(1 - r)^2 \leq \gamma_\epsilon(1 + r)^2.$$

Hence $r \geq (1 - \sqrt{\gamma_\epsilon})/(1 + \sqrt{\gamma_\epsilon})$. For this lower-bound value, (33) gives $\vartheta = r^2$ and makes $1 - \omega a + \vartheta$ range from $2r$ to $-2r$. Both roots therefore have modulus at most r , proving optimality. $\square$

A one-parameter momentum curve. For a fixed gap, minimax tuning equioscillates the endpoint modes, which imposes the linear relation

$$2(1 + \vartheta) = \omega(1 + \gamma_\epsilon), \quad \text{or equivalently} \quad \vartheta = \frac{1 + \gamma_\epsilon}{2}\omega - 1. \quad (35)$$

At the optimum, a more convenient one-parameter representation is

$$\vartheta = r^2, \quad \omega = (1 + r)^2, \quad \gamma_\epsilon = \left(\frac{1 - r}{1 + r}\right)^2, \quad 0 \leq r < 1. \quad (36)$$

Thus selecting r is equivalent to selecting an estimated gap and automatically couples the two algorithmic parameters. This reduces parameter search to one dimension. It does not remove the need for safeguards: an overestimate of γ_ϵ can produce parameters that are unstable for the true spectral interval.

For $\gamma_\epsilon \ll 1$, scalar extrapolation changes $1 - \gamma_\epsilon$ into $1 - 2\gamma_\epsilon + O(\gamma_\epsilon^2)$ and therefore preserves the scaling of the iteration count. In contrast, $r_{\text{mom}} = 1 - 2\sqrt{\gamma_\epsilon} + O(\gamma_\epsilon)$.

Over-relaxing the two block updates separately gives the nonlinear SOR variant studied in [18, 13]. Its optimal local parameter is

$$\omega_{\text{SOR}} = \frac{2}{1 + \sqrt{\gamma_\epsilon}},$$

and it has the same factor (34). Block SOR and the outer momentum step (30) are different algorithms, despite their identical optimal local rate. Their global convergence requires safeguards; the displayed parameters are local spectral prescriptions, not unconditional global guarantees.

Estimating the gap from iterates. The optimal parameters depend on an unknown local quantity. During an ordinary Sinkhorn run, set

$$d^{(\ell)} := \text{cent}_\beta(g^{(\ell)} - g^{(\ell-1)}).$$

Near the fixed point, $d^{(\ell+1)} = T_\epsilon^* T_\epsilon d^{(\ell)} + o(\|d^{(\ell)}\|)$, so the Rayleigh quotient

$$\hat{r}_\ell := \frac{\langle d^{(\ell)}, d^{(\ell+1)} \rangle_\beta}{\|d^{(\ell)}\|_\beta^2} \quad (37)$$

converges generically to σ_ϵ^2 when a dominant mode is present. The norm ratio $\|d^{(\ell+1)}\|_\beta / \|d^{(\ell)}\|_\beta$ is a simpler, slightly noisier alternative. In practice one clips $\hat{r}_\ell$ to $[0, 1)$, sets $\hat{\gamma}_\ell = 1 - \hat{r}_\ell$, and activates acceleration only after the estimate has stabilized. This also explains why extrapolation based on early iterates can be unreliable: the asymptotic spectral mode has not yet emerged.

4 Semi-dual variable projection

Eliminating one potential exactly converts the same operator $T_\epsilon^* T_\epsilon$ from an iteration Jacobian into a Hessian correction. This is the variable-projection mechanism: the poorly conditioned block coupling is removed by a Schur complement rather than handled by alternating block maximization.

This elimination is a special case of the variable-projection method of Golub and Pereyra [8, 9]. In general, if one block is optimized exactly (or minimized for the negative objective) and

the positive Hessian at a stationary point is $H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}$ with $H_{11} \succ 0$, then the reduced Hessian is

$$H_{\text{red}} = H_{22} - H_{21}H_{11}^{-1}H_{12}. \quad (38)$$

Its variational characterization gives, with extremal spectral values in infinite dimension, $\lambda_{\min}(H) \leq \lambda_{\min}(H_{\text{red}})$ and $\lambda_{\max}(H_{\text{red}}) \leq \lambda_{\max}(H)$, so exact projection cannot worsen the local spectral condition number after the gauge has been removed. General variable-projection theory does not, however, quantify this gain without additional information on the blocks. Here both *diagonal* blocks of the normalized full-dual Hessian are the identity, while the off-diagonal blocks are T_ϵ and T_ϵ^* . This special structure makes the reduced operator $I - T_\epsilon^*T_\epsilon$ and its conditioning explicitly computable from σ_ϵ .

Reduced objective and Hessian. For each g , the exact source-potential maximizer is $g^{\bar{c},\epsilon}$ from (15). The resulting concave semi-dual is

$$\mathcal{E}_\epsilon(g) := \mathcal{D}_\epsilon(g^{\bar{c},\epsilon}, g) = \int_{\mathcal{X}} g^{\bar{c},\epsilon} d\alpha + \int_{\mathcal{Y}} g d\beta. \quad (39)$$

Let

$$d\pi_g(x, y) := \exp\left(\frac{g^{\bar{c},\epsilon}(x) + g(y) - c(x, y)}{\epsilon}\right) d\alpha(x) d\beta(y), \quad (40)$$

and denote its second marginal by β_g . Its first marginal is exactly α .

Proposition 4.1 (Semi-dual derivatives and Schur complement). *For bounded directions k, k' on $\mathcal{Y}$,*

$$D\mathcal{E}_\epsilon(g)[k] = \int_{\mathcal{Y}} k d(\beta - \beta_g), \quad (41)$$

$$D^2\mathcal{E}_\epsilon(g)[k, k'] = -\frac{1}{\epsilon} \int_{\mathcal{X}} \text{Cov}_{\pi_g(\cdot|x)}(k(Y), k'(Y)) d\alpha(x). \quad (42)$$

At $g = g^*$, the positive Riesz curvature operator on $L^2(\beta)$ is

$$H_{\text{sd}} := -\nabla^2\mathcal{E}_\epsilon(g^*) = \frac{1}{\epsilon}(I - T_\epsilon^*T_\epsilon). \quad (43)$$

The identity extends from bounded directions to $L^2(\beta)$ by density. It is exactly the Schur complement of the f block in (12).

Proof. The envelope theorem gives (41). Differentiating the conditional Gibbs law in (40) gives the conditional covariance formula (42). At the optimum,

$$\int_{\mathcal{X}} \text{Cov}(k, k' | X) d\alpha = \langle k, k' \rangle_\beta - \langle T_\epsilon k, T_\epsilon k' \rangle_\alpha = \langle k, (I - T_\epsilon^*T_\epsilon)k' \rangle_\beta,$$

which proves (43). Finally, the Schur complement of I in $\begin{pmatrix} I & T_\epsilon \\ T_\epsilon^* & I \end{pmatrix}$ is $I - T_\epsilon^*T_\epsilon$, which is (38) in the present geometry. $\square$

Conditioning and first-order rates. The spectral gap that controls Sinkhorn is exactly the strong concavity of $\mathcal{E}_\epsilon$, or equivalently the strong convexity of $-\mathcal{E}_\epsilon$.

Proposition 4.2 (Semi-dual conditioning and optimization rates). *On $L_0^2(\beta)$,*

$$\frac{\gamma_\epsilon}{\epsilon} I \preceq H_{\text{sd}} \preceq \frac{1}{\epsilon} I, \quad \kappa_{\text{sd}} \leq \frac{1}{\gamma_\epsilon}. \quad (44)$$

Gradient ascent on the local quadratic model of $\mathcal{E}_\epsilon$, with optimal constant step $2\epsilon/(1 + \gamma_\epsilon)$, has factor

$$r_{\text{GD}} \leq \frac{1 - \gamma_\epsilon}{1 + \gamma_\epsilon}. \quad (45)$$

Heavy-ball or Chebyshev acceleration has factor

$$r_{\text{HB}} \leq \frac{1 - \sqrt{\gamma_\epsilon}}{1 + \sqrt{\gamma_\epsilon}}. \quad (46)$$

Proof. The spectrum of $T_\epsilon^* T_\epsilon$ on centered functions lies in $[0, 1 - \gamma_\epsilon]$, so the spectrum of (43) lies in $[\gamma_\epsilon/\epsilon, 1/\epsilon]$. This proves the curvature and condition-number bounds. The standard minimax polynomials on an interval give (45) and (46). $\square$

Since $\gamma_\epsilon = (1 - \sigma_\epsilon)(1 + \sigma_\epsilon)$, this condition-number bound is smaller than the doubly centered full-dual condition number in (13) by the factor $(1 + \sigma_\epsilon)^2$.

Quasi-Newton methods. In a finite discretization, $-\mathcal{E}_\epsilon$ is a smooth unconstrained convex problem after fixing its additive gauge. BFGS attempts to learn the inverse of $I - T_\epsilon^* T_\epsilon$ from gradient differences, precisely the operator responsible for the slow modes. Under the standard local smoothness and positive-definiteness assumptions, full BFGS is locally superlinear; limited-memory BFGS trades that guarantee for lower memory and often strong practical performance [14].

This strategy has a substantial OT history. Cuturi and Peyré used smooth dual and semi-dual formulations with L-BFGS for variational Wasserstein problems and developed the general regularized semi-dual viewpoint [4, 5]; stochastic first-order optimization of semi-duals was developed by Genevay et al. [10]. Khamlich, Romor, and Rozza use L-BFGS as the optimization core of a modern entropy-regularized semi-discrete solver, together with truncation, multilevel continuation, and fast geometric queries [11]. Qiu, Yin, and Wang also report and analyze L-BFGS on the fully discrete entropic dual as an alternative to long Sinkhorn runs [17].

These methods do not make the small- ϵ conditioning disappear: the inverse Hessian still grows like $1/\gamma_\epsilon$. Their advantage is that curvature information is learned and reused. Each quasi-Newton step is more expensive than one scaling update, so comparisons should count kernel evaluations or matrix-vector products rather than iterations. The semi-dual is especially attractive when one block can be eliminated analytically or integrated accurately, as in semi-discrete transport.

Remark 4.3 (Extrapolated Sinkhorn versus semi-dual gradient ascent). These methods are not identical globally. For the coupling π_g in (40), let $r_g := d\beta_g/d\beta$. The scaling identities give

$$r_g = \exp\left(\frac{g - \mathcal{S}_\epsilon(g)}{\epsilon}\right), \quad \nabla \mathcal{E}_\epsilon(g) = 1 - r_g, \quad \mathcal{S}_\epsilon(g) - g = -\epsilon \log r_g. \quad (47)$$

Thus a Sinkhorn cycle uses the logarithmic marginal defect, whereas semi-dual gradient ascent uses the additive defect. They agree only to first order near $r_g = 1$, where

$$\mathcal{S}_\epsilon(g) - g = \epsilon \nabla \mathcal{E}_\epsilon(g) + o(\|g - g^*\|). \quad (48)$$

Consequently, relaxed Sinkhorn with parameter ω is locally semi-dual gradient ascent with step $\epsilon\omega$, and (30) is locally its heavy-ball counterpart with momentum ϑ . This explains the agreement between (32)–(34) and (45)–(46), without claiming a global identity. Nesterov acceleration has the same square-root dependence on γ_ϵ [15], although its standard worst-case constant differs from the stationary heavy-ball factor.

Operator summary. All local statements in the note follow from one contraction:

Object	Operator at the optimum	Consequence
Full dual Hessian	$\epsilon^{-1} \begin{pmatrix} I & T_\epsilon \\ T_\epsilon^* & I \end{pmatrix}$	coupled conditioning
Sinkhorn full-cycle Jacobian	$T_\epsilon^* T_\epsilon$	local factor σ_ϵ^2
Semi-dual Hessian	$\epsilon^{-1}(I - T_\epsilon^* T_\epsilon)$	curvature γ_ϵ/ϵ
Accelerated local polynomial	$I - T_\epsilon^* T_\epsilon$	rate governed by $\sqrt{\gamma_\epsilon}$

Thus the slow convergence of Sinkhorn and the ill-conditioning of the semi-dual are complementary descriptions of the same near-deterministic correlation modes. Variable projection removes one coupled block; extrapolation changes the first-order polynomial; momentum and quasi-Newton methods approximate the inverse curvature more aggressively.

References

- [1] L. Chizat, A. Delalande, and T. Vaškevičius, Sharper exponential convergence rates for Sinkhorn’s algorithm in continuous settings, *Mathematical Programming*, 215:809–858, 2026. DOI: 10.1007/s10107-025-02242-z.
- [2] G. Conforti, A. Durmus, and G. Greco, Quantitative contraction rates for Sinkhorn’s algorithm: beyond bounded costs and compact marginals, *Annals of Applied Probability*, to appear, 2026. arXiv:2304.04451.
- [3] M. Cuturi, Sinkhorn distances: Lightspeed computation of optimal transport, in *Advances in Neural Information Processing Systems 26*, pages 2292–2300, 2013.
- [4] M. Cuturi and G. Peyré, A smoothed dual approach for variational Wasserstein problems, *SIAM Journal on Imaging Sciences*, 9(1):320–343, 2016. DOI: 10.1137/15M1032600.
- [5] M. Cuturi and G. Peyré, Semidual regularized optimal transport, *SIAM Review*, 60(4):941–965, 2018. DOI: 10.1137/18M1208654.
- [6] S. Eckstein, Hilbert’s projective metric for functions of bounded growth and exponential convergence of Sinkhorn’s algorithm, *Probability Theory and Related Fields*, 193:585–621, 2025. DOI: 10.1007/s00440-025-01366-9.
- [7] J. Franklin and J. Lorenz, On the scaling of multidimensional matrices, *Linear Algebra and its Applications*, 114–115:717–735, 1989.
- [8] G. H. Golub and V. Pereyra, The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate, *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973. DOI: 10.1137/0710036.
- [9] G. H. Golub and V. Pereyra, Separable nonlinear least squares: The variable projection method and its applications, *Inverse Problems*, 19(2):R1–R26, 2003.
- [10] A. Genevay, M. Cuturi, G. Peyré, and F. Bach, Stochastic optimization for large-scale optimal transport, in *Advances in Neural Information Processing Systems 29*, 2016.
- [11] M. Khamlich, F. Romor, and G. Rozza, Efficient numerical strategies for entropy-regularized semi-discrete optimal transport, *Computer Methods in Applied Mechanics and Engineering*, 453:118821, 2026. DOI: 10.1016/j.cma.2026.118821.
- [12] P. A. Knight, The Sinkhorn–Knopp algorithm: Convergence and applications, *SIAM Journal on Matrix Analysis and Applications*, 30(1):261–275, 2008. DOI: 10.1137/060659624.

- [13] T. Lehmann, M.-K. von Renesse, A. Sambale, and A. Uschmajew, A note on overrelaxation in the Sinkhorn algorithm, *Optimization Letters*, 16(8):2209–2220, 2022. DOI: 10.1007/s11590-021-01830-0.
- [14] J. Nocedal and S. J. Wright, *Numerical Optimization*, second edition, Springer, 2006.
- [15] Y. Nesterov, A method for solving the convex programming problem with convergence rate $O(1/k^2)$, *Soviet Mathematics Doklady*, 27:372–376, 1983.
- [16] B. T. Polyak, Some methods of speeding up the convergence of iteration methods, *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964.
- [17] Y. Qiu, H. Yin, and X. Wang, Efficient, stable, and analytic differentiation of the Sinkhorn loss, ICLR submission, OpenReview, 2023. OpenReview: uATOkwOZaI.
- [18] A. Thibault, L. Chizat, C. Dossal, and N. Papadakis, Overrelaxed Sinkhorn–Knopp algorithm for regularized optimal transport, *Algorithms*, 14(5):143, 2021. DOI: 10.3390/a14050143.